The Impact of Market Consolidation on Ticket Prices

Handong Zhang

10-4-2026

Table of Contents

I. Introduction....4

a. General description of industry....4

b. Motivation....4

c. Overview of industry background....4

d. Porter's Five Forces....5

e. Description of narrow analysis and findings....6

f. Organization of the paper....6

II. Institutional Background....6

a. Description of the product....6

b. Description of producers....7

c. Demand—Buyers....8

d. Cost—Suppliers....9

i. First tiers....9

ii. Second tiers....12

e. Government regulations impacting the industry....15

f. Historical development of industry....16

III. Porter's Five Forces....17

a. Entry costs and barriers.....17

b. Buyer bargaining power.....19

c. Substitute products.....20

d. Supplier bargaining power.....21

e. Nature of interfirm rivalry.....23

IV. Collusion history.....25

V. Narrow analysis.....30

a. Data sources and processing.....30

b. Difference in difference using low-concentrated routes as a control group...32

i. Factor inputs of the passenger airline industry.....32

c. Summary statistics of data.....32

d. Econometric model.....33

e. Enumeration of all maintained hypotheses.....34

f. Estimation methodology.....35

g. Estimates.....36

VI. Conclusion.....37

a. Summary of Findings.....37

b. Limitations and potential future research.....37

VII. References.....39

I. Introduction

a. General Description of Industry

The U.S. domestic passenger airline industry provides air transportation services across cities within the United States. The market is highly concentrated, with a few major carriers such as Delta Air Lines, American Airlines, and United Airlines accounting for the majority of the market share[1]. The industry is also characterized by high fixed costs, including aircraft investment, labor, and jet fuel expenses.

b. Motivation

My interest in this market stems from my personal background: as an international student, I traveled by air several times a year. Over time, I observed that the price of air tickets of major agencies has experienced a sharp increase in a similar pattern, and it remains high even though the price of fuel is stable. These experiences led me to question how prices in airline transportation markets are determined and whether the market is competitive. Specifically, I am interested in whether price movements fully reflect cost changes and competitive pressures, or whether market concentration facilitates coordinated pricing behavior.

c. Overview of Industry Background

Following the Airline Deregulation Act, most government controls over pricing and route entry were removed[12]. Since then, the industry has undergone substantial consolidation

through a series of mergers, including the Delta–Northwest merger, the United–Continental merger, the Southwest–AirTran merger, and the American–US Airways merger.[24] These mergers have increased market concentration and may have facilitated more strategic pricing behavior among airlines.

d. Porter's Five Forces

Using Porter's Five Forces, the competitive structure of the U.S. domestic airline industry can be summarized as follows. Buyers consist of a large number of individual consumers and a smaller segment of travel agencies[2]. Travel agencies typically negotiate fares with airlines and may share markups with carriers[19], while individual passengers generally behave as price takers. Suppliers operate across two tiers, including airport operations and aircraft maintenance, repair and overhaul (MRO) services at the first tier, and aircraft and parts manufacturing at the second tier, with relatively high entry barriers in both segments[3,4,7].

The threat of substitutes is limited, as there are no close alternatives for air transportation over medium- and long-distance travel. The threat of new entrants is also low, driven by structural and institutional barriers that restrict market entry. Finally, competitive rivalry among incumbent airlines is intense, as firms compete through pricing, route allocation, and flight frequency decisions within hub-and-spoke network structures[14]. Access to airport facilities and slots is also limited and often strategically constrained, further reinforcing competition among existing carriers[17].

e. Description of Narrow Analysis and Findings

This paper employed the difference-in-difference framework to analyze the impact of market consolidation on ticket prices using the DB1B dataset. The treatment group consists of routes that were originally moderately concentrated, but became highly concentrated after the consolidation, and the control group includes the routes that were always moderately- or low-concentrated. The post time period was defined as 2010 Q1, when Delta Air Lines took full control of Northwest Airlines[25]. The empirical results indicate that market consolidation led to an increase in average airfares on affected routes relative to the control group, suggesting that increased market power following consolidation may have contributed to higher prices.

f. Organization of the Paper

This paper first discusses the institutional background of the U.S. airline industry, followed by a Porter's Five Forces analysis of the competitive structure. It then introduces the empirical framework and narrow analysis based on a difference-in-differences approach. Finally, the paper reports the main findings and concludes.

II. Institutional Background

a. Description of the product

The U.S. domestic airline industry can be broadly divided into four main segments: mainline passenger transportation, cargo transportation, regional passenger transportation, and other related services. Mainline passenger transportation represents the largest portion of the industry, accounting for 46.2% of total revenue ($112.7 billion). This segment primarily consists of major airlines operating large aircraft on medium- and long-haul routes across the United States. Cargo transportation accounts for 22.5% of industry revenue ($54.9 billion) and includes the shipment of goods and freight by air. Regional passenger transportation represents 22.1% ($53.9 billion) and is mainly provided by smaller carriers that operate shorter routes and connect smaller airports to major airline hubs. The remaining 9.2% ($22.4 billion) consists of other related activities within the airline industry[1].

b. Description of Producers

The U.S. domestic airline market is primarily served by several major carriers. According to the latest data from IBISWorld, United Airlines Holdings, Inc. accounts for approximately 22.6% of the market with revenues of $55.1 billion, followed by Delta Air Lines, Inc. with 21.6% ($52.6 billion), and American Airlines Group Inc. with 20.6% ($50.1 billion). Southwest Airlines Co. occupies around 10.3% of the market, generating approximately $25.2 billion in revenue. The remaining 24.9% of the market, equivalent to $60.8 billion, is shared among a number of smaller airlines.

Overall, the industry is characterized by the presence of several large carriers with relatively

comparable market shares rather than a single dominant firm. No individual airline holds a sufficiently large share of the market to exercise dominant firm behavior. Instead, the market structure more closely resembles an oligopoly in which a small number of major firms collectively account for the majority of industry output.

Such a market structure may create conditions conducive to strategic interaction among firms. In particular, when a small number of producers repeatedly compete in the same markets, there may be incentives to coordinate pricing behavior in order to stabilize prices or avoid aggressive competition. Furthermore, industry data from the past five years indicate that the market shares of the major airlines have remained relatively stable, which may reflect the persistence of competitive positions within the industry[1].

c. Demand--Buyers

Demand in the U.S. domestic passenger airline industry primarily comes from two major categories of buyers: travel agencies and individual consumers. According to data reported by IBISWorld, travel agencies account for only a relatively small share of total demand, approximately $8.3 billion in airline service purchases[2]. These agencies typically act as intermediaries, facilitating ticket bookings and travel arrangements on behalf of customers rather than directly consuming airline services themselves.

In contrast, individual consumers constitute the dominant source of demand in the market for business or leisure travel[1]. This group includes travelers who purchase airline tickets directly from carriers or through online booking platforms for personal or business travel within the United States. As a result, the majority of airline service purchases in the domestic market are made by individual customers rather than institutional intermediaries[1].

d. Cost--Suppliers

Costs in the U.S. domestic passenger airline industry primarily come from two tiers: the first tier includes airport operations in the US; aircraft maintenance, repair & overhaul in the US; gasoline & petroleum wholesaling in the US; and aircraft, marine & railroad transportation equipment wholesaling in the US. The second tier includes aircraft, engine & parts manufacturing in the US; soda Production in the US; snack food production in the US; and bottled water production in the US[1].

i. First Tiers

The upstream industries can be discussed in sequence. The first component is airport operations in the United States. Airport operations are largely concentrated in regions with large populations and strong economic activity. According to industry data, states such as Florida (10.3%), Texas (9.6%), and California (9.1%) account for the largest shares of airport establishments[3]. This distribution reflects factors such as strong tourism demand, large

population bases, and the presence of major metropolitan areas, including cities such as Los Angeles and San Francisco.

Other regions also play an important role in supporting domestic air transportation. For example, the Southwest and Great Lakes regions host several major aviation hubs that facilitate domestic connectivity and business travel. States such as Illinois (3.1%), Ohio (2.5%), and Michigan (2.3%) represent important aviation centers within the Great Lakes region. In addition, the Mid-Atlantic region benefits from dense population centers and significant political and financial activity in cities such as New York and Washington, D.C., which support extensive airport operations and high levels of passenger traffic[3].

The second component of the upstream supply chain is the Aircraft Maintenance, Repair, and Overhaul (MRO) market in the United States. This market is characterized by the presence of several major service providers alongside a large number of smaller firms. According to industry data, General Electric Company holds approximately 19.1% of the market, generating about $5.7 billion in revenue. The Boeing Company accounts for roughly 11.5% of the market with revenues of $3.4 billion, while AAR Corp. represents around 3.4%, with revenues of approximately $999.7 million. The remaining 66.0% of the market, equivalent to about $19.7 billion, is shared among numerous smaller companies that provide maintenance, repair, and overhaul services to airlines and other aircraft operators[4].

The third upstream component is the gasoline and petroleum wholesaling market in the United States, which supplies jet fuel to the airline industry. Within this broader market, jet fuel accounts for approximately 4.7% of total industry sales. The market itself is relatively fragmented and includes a large number of suppliers. For example, BP plc holds about 5.4% of the market with revenues of approximately $42.7 billion, while the remaining 94.6% of the market, equivalent to $749.2 billion, is distributed among many other producers and wholesalers[5].

The final component of the first tier is the aircraft, marine, and railroad transportation equipment wholesaling industry in the United States. This industry supplies a variety of transportation-related equipment and parts, with aircraft equipment and supplies, including engines and components, representing the largest product category. This segment accounts for approximately $30.4 billion, or 51.4% of total industry revenue, constituting more than half of the market. New aircraft represent the second-largest category, accounting for $16.9 billion, or 28.6% of total industry revenue. Other segments include marine machinery, equipment, and supplies, which account for about $4.5 billion (7.6%), while the remaining 12.3% consists of other transportation equipment and related products[6].

Overall, the industry is highly fragmented and consists primarily of small- to medium-sized firms. Many companies specialize in the distribution of used aircraft parts and components. Nevertheless, several larger distributors also operate in the market. For example, Boeing

Global Services (formerly Aviall) is a major distributor of aerospace parts, and its sales account for approximately 20% of the total U.S. market[6].

ii. Second Tiers

The first component in the second tier is the aircraft, engine, & parts manufacturing in the US. Different from the aircraft, marine, and railroad transportation equipment wholesaling industry in the United States, the former industry focuses on producing aircraft, engines, and related components, while the latter primarily distributes and wholesales transportation equipment and parts rather than producing them. In the U.S. market, the production of aircraft accounts for more than half of total industry revenue, generating approximately $155.5 billion[7]. The remaining revenue is derived from aircraft engines and engine parts, as well as other aircraft components and auxiliary equipment, which together constitute the rest of the market[7].

The industry is characterized by the presence of several major firms, including Boeing, RTX Corporation, GE Aviation, and Lockheed Martin, in addition to numerous medium- and small-sized companies. These firms account for approximately 19.7%, 18.7%, 11.2%, and 41.2% of the market share, respectively[7].

Additionally, soda, snack food and bottled water production also accounts for the second-tier

suppliers, while these are not essential to the analysis below[1].

According to the data from the Airlines for America(2026), the first tier of suppliers consists of industries that provide essential inputs directly required for airline operations. Airport operators supply critical infrastructure, including runways, terminals, and gate access, and the associated expenditures are largely captured under transport-related services, which account for 10.4% of total operating expenses according to the Passenger Airline Cost Index (PACI, 2025). Aircraft maintenance, repair, and overhaul (MRO) providers ensure that aircraft remain airworthy and compliant with safety regulations; these expenditures are reflected in maintenance materials, which represent approximately 2.0% of operating costs. Fuel suppliers constitute another key input, with fuel expenses accounting for 16.6% of total operating costs, making fuel the second-largest cost component after labor. In addition, aircraft and transportation equipment wholesalers supply aircraft and related capital equipment through procurement and leasing arrangements. These costs are captured under aircraft rents and ownership, which account for approximately 5.2% of operating expenses[8].

U.S. Passenger Airline Cost Index (PACI): 3Q 2025

	PACI Index (2000 = 100)	Index YOY	$ Expense	Share of Op Expense
COMPOSITE (Pre-Tax) EXPENSES*	219.3	1.4%	$63,015.0M	
Labor	266.4	5.8%	$21,864.4M	35.3%
Fuel	265.0	-7.2%	$10,278.2M	16.6%
Transport-Related	225.1	2.8%	$6,449.6M	10.4%
Professional Services	174.9	3.7%	$5,776.6M	9.3%
Aircraft Rents & Ownership	85.5	-1.2%	$3,225.6M	5.2%
Non-Aircraft Rents & Ownership	177.3	10.4%	$3,386.6M	5.5%
Employee Business Expenses	278.2	8.9%	$1,468.2M	2.4%
Landing Fees	278.3	6.6%	$1,457.9M	2.4%
Food & Beverage	112.1	5.8%	$1,388.7M	2.2%
Maintenance Material	154.0	-0.7%	$1,267.3M	2.0%
Passenger Commissions	15.0	-5.4%	$357.3M	0.6%
Communication	70.2	2.6%	$418.8M	0.7%
Utilities & Office Supplies	146.8	20.0%	$383.7M	0.6%
Advertising & Promotion	69.3	2.6%	$390.0M	0.6%
Non-Aircraft Insurance	102.1	-1.5%	$101.5M	0.2%
Aircraft Insurance	88.6	21.6%	$45.6M	0.1%
Other Operating Expense	154.9	-18.3%	$3,602.0M	5.8%
Interest	64.9	-0.1%	$1,153.1M	

The second tier of suppliers includes upstream manufacturing industries and supporting service providers whose inputs are indirectly incorporated into airline operations. Aircraft, engine, and parts manufacturers produce aircraft and major components used by airlines. Although these costs are not reported separately in airline operating expenses, they are indirectly reflected through aircraft ownership costs (5.2%) and maintenance-related expenditures (2.0%). In addition, several food and beverage industries—including soda production, snack food manufacturing, and bottled water production—supply products used in onboard passenger services. These inputs are reflected in food and beverage expenses, which account for approximately 2.2% of airline operating costs. Compared with core operational inputs such as fuel and airport services, these catering-related expenditures

represent a relatively small share of the overall airline cost structure[8].

e. Government Regulations Impacting the Industry

The U.S. domestic passenger airline industry is subject to federal regulations covering three main areas: aviation safety, economic and consumer protection, and airport security.

Aviation safety is supervised by the Federal Aviation Administration (FAA), which sets rules for aircraft certification, pilot licensing, operational procedures, and maintenance. These regulations apply to daily operations, routine inspections, and emergency protocols, requiring airlines to maintain high safety standards across all flights[9].

The U.S. Department of Transportation (DOT) monitored the economic conditions and made restrictions on the companies to protect consumers. The DOT regulates airline practices such as fare transparency, disclosure of flight delays, and passenger compensation policies. These rules help ensure that travelers have clear information, receive fair treatment, and can keep informed when making decisions regarding ticket purchasing or responses to service disruptions[10].

Airport security and passenger screening are handled by the Transportation Security Administration (TSA). The TSA establishes standards for baggage inspection, airport access

control, and overall security procedures[11].

Notably, the Airline Deregulation Act of 1978 significantly reshaped the industry by removing federal control over routes and fares. Prior to deregulation, the government directly influenced market entry and pricing decisions; after the Act, airlines gained the freedom to determine routes and set prices based on market conditions[12]. This policy adjustment increased competition and allowed firms to respond more freely to market conditions. At the same time, the greater flexibility in pricing and strategic interaction among airlines also changed the competitive landscape, creating market conditions that may incentivize coordinated pricing behavior and potential collusion.

f. Historical Development of Industry

The history of the U.S. domestic passenger airline industry can be broadly divided into three periods: the regulation era, the deregulation period, and the period of industry consolidation.

One milestone of the US passenger airline industry is the announcement of the Airline Deregulation Act in 1978. Before the issue of this act, the industry was overseen by Civil Aeronautics Board(CAB), the government. The government controlled market entry, route allocation, and airfare levels. Airlines were required to obtain government approval before they developed new routes, and ticket prices were largely determined by regulatory decisions

rather than market competition[12]. As a result, competition among airlines was relatively limited and primarily focused on service quality rather than price. As a result, competition among airlines was relatively limited at that time.

After the announcement of the Airline Deregulation Act, most government controls over pricing and route entry were removed, which allowed airlines to make their own operational and pricing decisions[12]. After deregulation, some new airlines entered the market, and existing carriers expanded their market share. During this period, many airlines adapted to the hub-and-spoke network system, which cut costs by reducing the number of direct routes required.[14]

The third period is characterized by industry consolidation. Beginning in the 2000s, a series of mergers significantly increased market concentration in the industry. Major mergers such as the Delta-Northwest merger in 2008, the United-Continental merger in 2009, the Southwest-AirTran merger in 2010, and the American-US Airways merger in 2013 reshaped the competitive landscape and resulted in a market dominated by a small number of large carriers[24].

III. Porter's Five Forces

a. Entry Costs and Barriers

Entry into the U.S. domestic passenger airline industry is constrained by several structural and institutional factors.

First, new entrants face high start-up costs and capital investment. To enter the market, one needs to purchase maintenance facilities, hangar space, and airport infrastructure while also investing in specialized equipment and safety systems[1]. Additionally, capital investment in purchasing the aircraft is also a particularly large cost for new entrants as there are a limited number of manufacturers that they can choose from. Although leasing aircraft may reduce the initial financial burden, capital requirements remain considerable.

Additionally, the industry must operate under extensive regulatory oversight by government agencies. For instance, safety and operational standards are primarily regulated by the Federal Aviation Administration (FAA), which is responsible for certifying airlines, approving aircraft operations, and issuing licenses for commercial pilots and flight crews[9]. In addition, economic regulation and consumer protection issues are overseen by the United States Department of Transportation (DOT), which reviews airline licensing, monitors unfair or deceptive practices, and evaluates certain cooperative agreements between airlines[10]. Competition-related issues, particularly mergers and potential collusive behavior, fall under the jurisdiction of the United States Department of Justice Antitrust Division, which investigates antitrust violations and reviews major airline mergers. Environmental and workplace standards are also enforced by agencies such as the Environmental Protection

Agency and the Occupational Safety and Health Administration.

What's more, even though the product itself seems homogeneous, the airlines make efforts to make their product differential. Larger airlines tend to offer better amenities such as a wider variety of snacks and drinks, and better recreation facilities along the trip. They also established the loyalty program, and they have partnerships with companies in different industries such as financial institutions and hotel industry. Finally, a proven safety record and a positive reputation can make the airline distinguished as well.

Last but not least, pilots are always in shortage in the passenger airline industry. It is costly to train a qualified pilot, and it can be extremely hard for start-ups to hire pilots. Moreover, airline personnel are in high demand as airlines try to meet increasing travel plans[5].

b. Buyers Bargaining Power

There are two main types of buyers in the U.S. domestic passenger airline market: travel agencies and individual customers.

The first category consists of travel agencies, which account for a smaller share of ticket sales compared with airlines' direct distribution channels. According to Vinod(2024), airlines and travel agencies often reach agreements on negotiated fares with markups. When a customer

purchases a private fare through a travel agency, the agency marks up the net fare negotiated with the airline and submits the ticket value to the airline through the Billing and Settlement Plan (BSP) for settlement[19]. In practice, this arrangement allows customers to obtain fares that may be lower than publicly available prices, while travel agencies earn markups and airlines can promote the sale of specific fares[19].

The second category of buyers consists of individual passengers who purchase tickets directly from airline websites. Since each customer typically buys only a very small number of tickets--often one or two per transaction--individual consumers have little ability to negotiate prices directly with airlines. In addition, loyalty programs, such as the MileagePlus program offered by United Airlines, encourage repeat purchases and reduce switching behavior. However, online travel platforms allow consumers to easily compare fares across airlines, which lowers switching costs and increases price transparency. As a result, passengers tend to be highly price sensitive, which intensifies price competition among airlines.

Overall, while passengers can easily compare prices and switch between airlines, the limited purchasing volume of individual consumers means their direct bargaining power over airlines remains relatively limited.

c. Substitute Products

According to a report from IBISWorld, there are no close substitutes for domestic air travel in the United States. However, railway transportation and personal driving may serve as potential alternatives in certain situations.

Rail transportation is often considered a viable option for short- and medium-distance trips. Nevertheless, compared with domestic airlines, rail travel generally offers more limited network coverage and longer travel times, particularly for long-distance journeys. In addition, the U.S. rail infrastructure is relatively underdeveloped compared with air transportation, which benefits from an extensive network of airports across the country.

Driving may also serve as an alternative for short-distance travel. However, airlines benefit from economies of scale that enable them to transport passengers over long distances at relatively low per-passenger costs. Furthermore, air travel is significantly faster, allowing passengers to travel between the East and West coasts within a few hours[5].

Therefore, while rail transportation and driving may substitute for air travel in short-distance trips, their limitations in speed and network coverage reduce their effectiveness as substitutes for long-distance domestic travel.

d. Suppliers Bargaining Power

The bargaining power of suppliers in the U.S. domestic passenger airline industry varies across different types of upstream industries.

First, aircraft manufacturing and equipment supply represent one of the most powerful segments of the supply chain. The number of firms capable of producing commercial aircraft and their major components is extremely limited. As a result, airlines have few alternatives when purchasing aircraft of comparable quality and performance. The high technological barriers to entry and the large capital requirements further limit competition among manufacturers, which gives aircraft producers strong bargaining power over airlines.

Second, airport operations also exhibit considerable supplier power. Establishing a new airport requires approval from the Federal Aviation Administration and compliance with extensive federal regulations. In addition, the construction of airports involves substantial upfront investment in land, infrastructure, advanced technology, and security systems. Airports also face significant operating costs, including labor and specialized equipment for refueling, ground handling, and maintenance services. These factors create high barriers to entry and limit the availability of alternative airport facilities within a given region.

Third, airlines rely heavily on jet fuel supplied through the petroleum industry. Because aircraft operations depend almost entirely on jet fuel, airlines are highly exposed to

fluctuations in oil prices and have very limited short-term substitutes for this input. This dependence gives fuel suppliers a certain degree of bargaining power over airlines.

Finally, food and beverage producers represent a relatively weak group of suppliers. These products account for only a small share of airline operating costs—approximately 2.2% of total costs—and airlines can easily switch among numerous suppliers in this market. As a result, food and beverage providers generally possess limited bargaining power.

Overall, supplier bargaining power varies across inputs, with aircraft manufacturers and airport operators holding stronger power, while food and beverage suppliers have relatively limited influence.

e. Nature of Interfirm Rivalry

According to the research of Barwick and Yuan (2024), airlines do not initially compete through pricing to attract consumers. Instead, they first make market entry decisions and determine flight frequencies. These decisions are typically organized under the hub-and-spoke network structure. Under this system, rather than establishing direct connections between every pair of cities, which would require $C(C-1)/2$ nonstop routes, the network requires only $C-1$ routes that connect cities through a central hub. In practice, this structure allows airlines to significantly reduce operating costs while maintaining extensive network coverage[14].

To establish dominance in the hubs, airlines would increase flight frequency in those airports: in earlier literature, Adib Ghobrial (1991) found a significant positive relationship between airline profitability and hub dominance, suggesting that airline profitability depends not only on the number of routes operated but also on the extent to which an airline dominates its hub airports[15]. Similarly, Severin Borenstein (1991) documented the existence of an "S-curve" effect at hub airports: airlines with larger flight shares at an airport tend to capture disproportionately larger passenger shares[16].

According to Ciliberto and Williams (2010), airlines may restrict competitors' access to airport facilities at their hub airports to maintain market dominance. This behavior is particularly common in their control over gate availability, leasing arrangements, and other operational constraints[17]. By occupying more gates than necessary, airline companies can limit entry and competition, thereby increase hub premiums and charging higher ticket fares.

What's more, according to the data from IBIS World, the top four players in the market occupied more than 75% of the market share. Additionally, those large airline operating companies, such as United Airlines Inc., have loyalty programs and frequent flyer benefits to foster customer loyalty. For example, Lederman (2008) shows that airlines use frequent flyer programs (FFPs) to strengthen customer loyalty by awarding points for flight purchases, which can be redeemed for discounts or rewards such as free tickets and class upgrades[18].

These programs typically structure minimum point thresholds before redemption is possible, which encourages repeated purchases from the same airline. Moreover, the reward structure often increases nonlinearly with the number of points required, which increases the stickiness of the program[18]. In addition, many FFPs include elite tiers that grant status to passengers who meet annual mileage requirements, providing benefits such as priority boarding and other privileges. Empirical evidence suggests that the presence of FFPs is associated with a 3.7–5% increase in ticket prices[18].

What's more, they partnership with major financial institutions by creating points by running their operations and selling these points to banks via co-branded credit cards. Banks then will award these points to cardholders based on their spending while earning profit from swipe fees. However, those financial institutions only have incentives to collaborate with large airline companies, which put small airlines in disadvantaged positions[1].

IV. Collusion History

According to the U.S. Department of Justice, Antitrust Division (1992), the first well-documented case of collusion in the U.S. passenger airline industry dates back to the 1980s: the Airline Tariff Publishing Co. Case. The case involved several major U.S. airlines, including Alaska Airlines, American Airlines, Continental Airlines, Delta Air Lines, Northwest Airlines, Trans World Airlines, and United Airlines[20,21].

Airline Tariff Publishing Company (ATP), a fare dissemination entity owned by the defendants, functioned as a centralized information exchange platform through which airlines filed and accessed fare data. The DOJ alleged that the defendants used ATP not merely for price publication, but as a mechanism to signal intended fare changes, eliminate discounts, coordinate fare increases, and set fare restrictions in various city pairs markets. By reading signals from other airlines, one could condition their own pricing decisions accordingly[20,21].

Furthermore, the ATP system operated as an internal enforcement structure. The defendants monitored the fare changes proposed by other airlines on ATP, including ticketing dates and links among different markets. The defendants linked the last ticket dates link city-pair markets using footnotes designators. This information exchange system lessened the uncertainty concerning others' pricing structures and enhanced the sustainability of collusion[20,21].

In re Air Cargo Shipping Services Antitrust Litigation (2007) involved allegations of a global cartel in the international airfreight market[22]. Although this case is not directly linked to the passenger airline industry, it is highly correlated, and the plus factors revealed in this case can help detect potential collusive behaviors in the passenger airline industry. Begin in the 1990s, many Airfreight carriers, trade groups, or other entities including Air Canada, AC Cargo LP,

Société Air France, Koninklijke Luchtvaart Maatschappij N.V, Aerolíneas Brasileiras S.A. d/b/a ABSA Cargo Airline, etc. were involved in the cartel[22].

In practice, the defendants were alleged to have participated in a coordinated scheme to fix, raise, maintain, and stabilize prices for Airfreight Shipping Services by posing fuel, security, war risk(insurance) and U.S. customs surcharges to customers, and refused to offer a discount concertedly, thereby facilitating coordinated price increases while limiting price competition[22]. In practice, these defendants would announce price-fixing policies using similar reasons.

In addition to surcharge harmonization, those conspiring entities reached an agreement on increase in yields simultaneously to increase revenue[22]. The conspiracy further involved customer allocation arrangements across international routes, effectively restricting competition and reducing shippers' access to competitive pricing.

To sustain the cartel, the defendants allegedly exchanged competitively sensitive information, including individual carrier yield data, which functioned as a monitoring mechanism to detect deviations. The complaint also asserted that participants concealed their conduct to delay detection and minimize legal exposure[22].

In the European Commission Decision of Case COMP/39.596 (2011), airline companies including British Airways Plc., American Airlines, Inc., and Iberia Líneas Aéreas de España, S.A. were alleged to set up a revenue-sharing joint venture that covers all passenger air transport services such as access to connecting passengers at hubs through standard industry interline and airport services on routes between Europe and North America by Virgin Atlantic Airways[23]. In the complaint, Virgin Atlantic alleged that the defendants reached an agreement on extensive scheduling, pricing coordination on routes between specific city-pair routes and revenue-sharing arrangements. In particular, Virgin Atlantic claimed that the agreement enabled the defendants to coordinate extensively on capacity and fares, leading to higher ticket prices, reduced service quality, and diminished incentives for innovation[23].

Even though this case is considered to have no longer grounds for action[23], it is worth noting that revenue-sharing joint ventures are not automatically treated as cartels, and this distinction is particularly relevant when distinguishing between collusion and legally permissible strategic alliances.

In *In re Domestic Airline Travel Antitrust Litigation* (2017), the four largest commercial air passenger carriers in the United States, American Airlines, Delta Air Lines, Southwest Airlines, and United Airlines, were accused of fixing, raising, maintaining, and stabilizing prices for passenger transportation services within the United States since 2009. The plaintiffs argued that despite a decline in jet fuel prices after 2009, ticket prices charged by these

airlines remained high compared with other carriers[24]. However, no final court decision determined whether the airlines had formed a cartel or engaged in coordinated pricing behavior.

According to the complaint, the defendants allegedly reduced capacity growth at their hub airports despite steadily increasing demand during the period[24]. The plaintiffs further claimed that the airlines increased prices on these routes compared with airlines that were not defendants. In some cases, the defendants were alleged to charge identical or nearly identical prices on certain routes. For example, the complaint cited a flight from Houston to New York City on August 13, 2015, where US Airways charged $575, while the defendant airlines charged $1,331, $1,467, and $1,467, respectively[24]. The complaint also alleged that the airlines imposed various ancillary fees, such as charges for checked bags and seat selection, which further increased the total cost for passengers[24].

Additionally, the plaintiffs argued that the airlines limited capacity expansion in their hub markets and avoided aggressive entry into competitors' hubs. This behavior allegedly reduced head-to-head competition and allowed dominant airlines at major hubs to maintain a large share of available seats[24].

Finally, the complaint stated that the airlines used the fare-filing platform operated by the

Airline Tariff Publishing Company (ATPCO) to monitor competitors' pricing and fare changes[24], which could potentially serve as a mechanism facilitating coordination among airlines.

V. Narrow Analysis

a. Data Sources and Processing

This paper used the DB1B database on the U.S passenger airline industry from the Bureau of Transportation Statistics, the producer price index of jet fuel and pilot wage statistics in the National Occupational Estimates Report from the Bureau of Labor Statistics.

$$HHI = 10000 \cdot \sum_{i=1}^{n} s_i^2 \quad (1)$$

In order to distinguish between the control and treatment groups, I construct the Herfindahl-Hirschman Index (HHI) at the carrier level using the standard formula (1) described above, where i is the index of carriers operating within a given route, and s_i denotes the corresponding market share of carrier i. The data are then aggregated to the market level by collapsing the dataset after computing the HHI. The sample period spans from 2008 Q1 to 2012 Q1.

The merger between Delta Air Lines Inc. and Northwest Airlines Inc., which occurred in 2008Q4, suggests a natural breakdown for a sample split[24]. However, it was not until 2010

Q1 that Delta Air Lines took full control of Northwest Airlines regarding the ticket prices, systems, and brands[25]. Therefore, 2010Q1, rather than 2008 Q4, is used as the cutoff period to distinguish between the pre-merger and post-merger regimes. All quarters prior to the merger are classified as the pre-merger period, while 2010Q1 and subsequent quarters are defined as the post-merger period.

Following the guidelines established by the U.S. Department of Justice (2010), markets with an HHI exceeding 2,500 are classified as highly concentrated. Therefore, routes became highly concentrated after the market consolidation are assigned to the treatment group, while those below are used as the control group[26].

The producer's price index of jet fuel is a monthly time series, and a three-month moving average of the index is constructed to smoothen the volatility of the data. Specifically, the three-month average from the previous quarter is used as the input for the current quarter in the regression. For example, the average over 2007Q4 is used as the regressor for 2008Q1.

The pilot wage data, obtained from the MIT Airline Data Project(2023), is reported as an annual time series. To match the quarterly frequency of the dataset, the annual values are carried forward within each year, assigning the previous year's value to all quarters of the current year. This approach implicitly assumes that pilot wages remain constant within a

given year.

b. Difference-in-Difference Using Low-Concentration Routes as a Control Group

i. Factor Inputs for the Passenger Airline Industry

According to Airlines for America (2026), labor and jet fuel costs jointly account for approximately 51.9% of total operating costs. Specifically, labor costs account for 35.3%, while jet fuel costs account for 16.6%[8].

In addition, evidence from the MIT Airline Data Project suggests that the average annual wages of pilots and co-pilots accounts for about half of the labor cost budgets, which highlights the importance of pilot wages as a key component of airline labor costs, and therefore is included as the regressor[27].

c. Summary Statistics of Data

Variable	Obs	Mean	Std. dev.	Min	Max
ln_fare	967,553	5.593902	.471801	2.995732	11.50707
HHI_10000	967,553	6118.866	3107.16	689.8344	10000
jetfuel	967,553	256.1843	62.41189	137.53	353.03
pilot_wage	967,553	132.0313	4.976174	125.858	139.372

Table (1)

This section presents summary statistics for the main variables used in the analysis. Table (1) reports the mean, standard deviation, and range of key variables, including rate of change for market fares, market concentration index(HHI), jet fuel price index and pilot wage index.

d. Econometric Model

The empirical analysis for this paper is built on the difference-in-difference framework, which is designed to estimate the difference in price increase caused by the change in market concentration because of the Delta–Northwest merger. The merger consolidated the market and resulted in different levels of competition structure.

The regression formula is defined as:

$$\ln(mktfare)_{i,t} = \beta_0 + \beta_1(post_t \cdot treat_i) + \beta_2 d_jetfuel_t + \beta_3 pilotwage_t + \varepsilon_{i,t} \quad (1)$$

In this specification, $\ln_fare$ denotes the log change in ticket prices at the route level, and $post_t$ is a dummy variable equal to one for observations after 2010Q1. $Treat_i$ is an indicator variable equal to 1 for routes whose HHI was below 2,500 before 2010Q1 but exceeded 2,500 after 2010Q1, and 0 otherwise. $Post_t \cdot treat_i$ is the interaction term, which captures

the difference-in-differences effect.

Control variables include $jetfuel_t$, which measured the three-month average producer price index in the previous quarter; its first difference $d_jetfuel_t$ is taken to eliminate potential time trends. Additionally, $pilot_wage_t$ denotes pilot wages from the previous year in the regression model. Finally, $\varepsilon_{i,t}$ represents unobserved shocks.

e. Enumeration of all Maintained Hypotheses

Assumption 1: The control group is not affected by the treatment

The control group consists of moderately- or low-concentrated routes (HHI < 2500), which are served by multiple carriers, including network airlines, hybrid carriers, and ultra-low-cost carriers (ULCCs) [19]. The presence of multiple competitors makes coordinated pricing behavior challenging and less likely. Therefore, it is reasonable to assume that these routes are not influenced by the treatment, and this assumption holds.

Assumption 2: No change in the nature of the product

Following the assumption, the quality and nature of airline services for control group and treatment group should remain stable over time. There were no significant innovations in in-flight services or amenities during the period under study that would affect one group more

than the other. Thus, this assumption holds.

Assumption 3: No differential productivity shocks from technological change

There are no differential impacts on productivity between treatment and control groups arising from exogenous technological innovations. During the period of study, there were no pioneering breakthroughs in aircraft manufacturing technology that would affect one group more than the other. Therefore, this assumption holds.

Assumption 4: Symmetric impact of macroeconomic shocks

According to the assumption, exogenous shocks should affect both treatment and control groups similarly. During the Great Recession, the U.S. airline industry experienced a significant increase in jet fuel prices. This shock impacted all routes simultaneously, rather than targeting specific routes. Hence, it is reasonable to assume that both groups were affected symmetrically.

f. Estimation Methodology

The original dataset contains more than 900,000 observations, which inevitably contains a large number of outliers and inevitably undermines the explanatory power and performance of the regression model. To address this issue, observations with missing values were

removed, along with implausible fare values (e.g., market fares below $25). After this data cleaning process, approximately 25% of the observations were selected from the original sample.

This sampling strategy ensures that observations are randomly drawn within each period and across groups, which helps preserve the randomness assumption for OLS estimation. Meanwhile, it reduces the influence of outliers and prevents them from negatively affecting the explanatory power of the model.

g. Estimates

	(1)
	ln_fare
Post × Treat	0.289***
	(0.0786)
d_jetfuel	0.000834**
	(0.000359)
pilot_wage	-0.0241***
	(0.00812)
_cons	8.581***
	(1.040)
Observations	511
R-squared	0.0485

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

According to the regression output, the coefficient of the interaction term is positive and statistically significant at 1% significance level, which suggests that treated routes have a 28% increase in ticket price compared to the control group after industry consolidation in 2010 Q1.

VI. Conclusion

a. Summary of Findings

To sum up, this paper investigates the impact of market concentration resulting from mergers on airline ticket prices on airline ticket prices using a difference-in-differences approach. The analysis used data from the DB1B database from the Bureau of Transportation Statistics, producer price index of jet fuel and pilot wage statistics from the Bureau of Labor Statistics.

The estimation result indicates that the difference-in-difference effect(the interaction term) is positive and statistically significant at 1% significance level. The coefficient of the interaction term suggests that highly concentrated routes have a 28% more in fare increase compared to those moderately or low concentrated routes after the market consolidation in 2010 Q1. This finding is consistent with the idea that higher market concentration may reduce competition and allow airlines to coordinate and charge a higher price.

b. Limitations and Potential Future Research

Given the time constraints, this paper did not find an additional control group. The absence of a second control group may weaken the robustness of the findings, as the results rely on a single comparison between treated and control routes.

Additionally, the DB1B dataset represents only a 10% sample of routes per quarter, which may introduce selection bias. Furthermore, the reported airline prices contain implausible observations and discounted fares, and even after data cleaning, it is not possible to fully eliminate all outliers.

If additional time were available, this difference-in-difference framework would be used to estimate the effect of another market consolidation effect, such as the American-US Airways merger in 2013, to ensure the robustness of the model. The comparison across different time periods would provide stronger evidence on whether the observed effects are consistent.

VII. Reference

[1] Companies - 48111B Domestic Airlines in the US - MyIBISWorld

[2] At a Glance - 56151 Travel Agencies in the US - MyIBISWorld

[3]At a Glance - 48811 Airport Operations in the US - MyIBISWorld

[4]Performance - 48819 Aircraft Maintenance, Repair & Overhaul in the US - MyIBISWorld

[5]At a Glance - 42472 Gasoline & Petroleum Wholesaling in the US - MyIBISWorld

[6]At a Glance - 42386 Aircraft, Marine & Railroad Transportation Equipment Wholesaling in the US - MyIBISWorld

[7] https://my-ibisworld-com.ezaccess.libraries.psu.edu/us/en/industry/33641a/at-a-glance

[8] A4A Passenger Airline Cost Index (PACI) | Airlines For America

[9] Aviation Safety (AVS) | Federal Aviation Administration

[10] Aviation Consumer Protection | US Department of Transportation

[11] Mission | Transportation Security Administration

[12] S.2493 - 95th Congress (1977-1978): Airline Deregulation Act of 1978 | Congress.gov | Library of Congress

[13] United States Department of Justice Antitrust Division — Grokipedia

[14] Yuan, Zhe, and Panle Jia Barwick. "Network competition in the airline industry: An empirical framework." The RAND Journal of Economics (2024). Network Competition in the Airline Industry: An Empirical Framework | NBER

[15] Ghobrial, Atef. "Competition for hub dominance: some implications to airline

profitability and enplanement share." Journal of Aviation/Aerospace Education & Research 2.1 (1991): 5. https://commons.erau.edu/jaaer/vol2/iss1/5/

[16] Borenstein, Severin. "The dominant-firm advantage in multiproduct industries: Evidence from the US airlines." The Quarterly Journal of Economics 106.4 (1991): 1237-1266.

https://www-jstor-org.ezaccess.libraries.psu.edu/stable/2937963?searchText=%2810.2307%2F2937963%29&searchUri=%2Faction%2FdoAdvancedSearch%3Fq0%3D10.2307%252F2937963%26f0%3Dall%26f1%3Dall%26acc%3Doff%26so%3Drel&ab_segments=0%2Fbasic_search_gsv2%2Fcontrol&refreqid=fastly-default%3A4c5776fbe8075a512a94ffa560bd6cbb&seq=25

[17] Ciliberto, F., & Williams, J. W. (2010). Limited access to airport facilities and market power in the airline industry. The Journal of Law and Economics, 53(3), 467-495. https://www.journals.uchicago.edu/doi/abs/10.1086/605725

[18] Lederman, M. (2008). Are frequent-flyer programs a cause of the “hub premium”?. Journal of Economics & Management Strategy, 17(1), 35-66. https://onlinelibrary.wiley.com/doi/full/10.1111/j.1530-9134.2008.00170.x

[19] Vinod, B. (2024). Scale Matters: GDS Air Shopping. In Mastering the Travel Intermediaries: Origins and Future of Global Distribution Systems, Travel Management Companies, and Online Travel Agencies (pp. 151-191). Cham: Springer Nature Switzerland. Scale Matters: GDS Air Shopping | Springer Nature Link

[20] U.S. Department of Justice, Antitrust Division. (1992, December 21). *Complaint: U.S. v. Airline Tariff Publishing Company, et al.* (Civil Action No. 92-2854). Retrieved from https://www.justice.gov/atr/case-document/file/483631/dl

[21] United States Department of Justice, Antitrust Division. (1992, December 21). *Competitive impact statement: U.S. v. Airline Tariff Publishing Company* (Civil Action No. 92-2854) [PDF]. U.S. Department of Justice. https://www.justice.gov/atr/case-document/file/483616/dl

[22] In re Domestic Airline Travel Antitrust Litigation, 1:06-cv-00830 (E.D.N.Y. 2006). https://www.govinfo.gov/content/pkg/USCOURTS-nyed-1_06-cv-00830/pdf/USCOURTS-nyed-1_06-cv-00830-0.pdf

[23] European Commission. (2011, June 20). *Commission decision in case COMP/39.596 – British Airways/American Airlines/Iberia : Decision rejecting Virgin Atlantic's complaint (non-confidential version)* (Document C(2011) 4505 final). Publications Office of the European Union. https://ec.europa.eu/competition/antitrust/cases/dec_docs/39596/39596_4997_5.pdf

[24] *In re Domestic Airline Travel Antitrust Litigation*, MDL No. 2656, Misc. No. 15-1404 (CKK), Memorandum Opinion, U.S. District Court for the District of Columbia, Oct. 28, 2016. https://psu.instructure.com/courses/2449288/files/folder/Complaints%20Info%20for%20Class?preview=189344110

[25] Delta Air Lines - Wikipedia

[26] U.S. Department of Justice & Federal Trade Commission. (2010). *Horizontal Merger Guidelines.* Washington, DC: U.S. Department of Justice and Federal Trade Commission. Retrieved from https://www.justice.gov/atr/file/810276/dl?inline#page=19.60

[27] MIT Airline Data Project. (n.d.). *Average annual wages and salaries - Pilot and co-pilot personnel*. MIT Global Airline Industry Program. Retrieved October 24, 2023, from https://web.mit.edu/airlinedata/www/Employees&Compensation.html